Effective use of Natural resource (Solow augmented model

Outline of the presentation

- Aim of the paper and research question
- Literature review and background
- The relationship between natural resource abundance and growth
- Model of natural resource and efficiency of natural resource use
- Empirical review of relevant data
- Conclusion

Aim of the Paper

- We seek to survey the popular result in the literature on “resource curse”, and postulate some facts about relationship natural resource abundance and growth.
- Through augmenting Romer’s Natural resource Solow growth model, incorporate the effective use of natural resource to decrease the drag caused by incorporating natural resource (i.e. natural resource curse).
- Lastly, argue that More natural resources push aggregate income down, when institutions are bad, while resources raise income, when institutions are good

Research Question

How does the quality of political institutions influence the impact of natural resource abundance on economic development in resource-rich countries?

Background

Natural resources affect growth through multiple channels:

- Capital accumulation and investment incentives
- Rent-seeking, corruption, and political capture
- Human capital formation and education
- Technological progress and productivity

The net effect depends critically on whether resource rents are **productively invested** or **politically misallocated**.

The relationship between natural resource abundance and growth (Literature Review)

The Resource Curse Hypothesis

Empirical studies frequently find a negative correlation between resource dependence and growth, attributing this to:

- Dutch disease effects
- Volatility of commodity prices
- Weak institutions and governance failures
- Reduced incentives for innovation and education

5.2 Conditional Resource Effects

More recent literature challenges the inevitability of the resource curse, showing that:

- Countries with strong institutions often benefit from resource abundance
- Education, bureaucratic quality, and transparency mitigate negative effects
- Resource wealth can accelerate growth when paired with effective policy frameworks

Model I/III

- Romer's Sustainable Solow growth model used a stylized model with finite resources; the conclusions of which depend crucially on some fairly arbitrary assumptions.
- Production with natural resources R
- Let R(t) be the amount depletable natural resources used up in production at time t
- The simple Cobb Douglas function Y(t)=F(Kt, Rt, At Lt)
- All factors cannot grow at same rate, because
- AL grows at n + g
- T grows at 0
- R grows at –b
- K grows at some as-yet-undetermined rate
- Steady state:

  - In the steady state, $\frac{\dot{Y}}{Y} = \frac{\dot{K}}{K} = g_Y^*$, so $g_Y^* = \alpha g_Y^* - \beta b + (1-\alpha-\beta-\gamma)(g+n)$ or

    $$g_Y^* = \frac{(1-\alpha-\beta-\gamma)(g+n) - \beta b}{1-\alpha}.$$

Model II/III

- Results of Solow growth model with natural resource (i.e. Resource curse)
- There is a growth drag introduced by the presence of natural resources and land in the production function.
  - Growth drag term is zero if beta = 0
  - Romer cites Nordhaus estimates that the growth drag may be ~0.25%
- Romer notes that Cobb-Douglas assumption that elasticity of substitution among factors is one may not be correct.
  - If elasticity of substitution is bigger than one, then it is relatively easy to get along without R and T as they become scarce relative to K and L.
  - If it is less than one, then it is difficult to substitute and the growth drag will be more severe

Model III/III: Solow sustainable model with effective use of natural resource.

Let B(t) be the effective use of natural resource in the model.

We have Y(t)= F(Kt, BtRt, AtLt)

where Kt is the capital,

BtRt is effective natural resource,

AtLt is effective labour.

Bt grows at rate gB i.e.B’(t)/B(t)= gB

Interpretation of gB

A,B,L,R grows at constant rate,

Empirical review

- Education and natural resource, Corruption and natural resources, Bureaucratic quality and natural resources.
- Natural resources are neither inherently a curse nor a blessing.
- Institutional quality determines the sign and magnitude of their growth effect.
- Efficiency in resource use is a crucial missing link in traditional growth models.

Conclusion

- This analysis reframes the resource curse as an institution-dependent phenomenon. By augmenting the Solow model to include effective resource use, the paper demonstrates that:
- Poor institutions transform resource wealth into economic drag
- Strong institutions enable natural resources to raise long-run income
- Policy reforms targeting governance, education, and efficiency are central to sustainable growth